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Performance evaluation of electro-mechanical railway interlocking system for digital twin application

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ABSTRACT

The industries engaged healthcare, manufacturing, and construction automated with exquisite creativity and innovation at the doorstep of Industry 4.0. In this process, digital twins are diligently used to supplement industrial processes, reduce costs, monitor assets, streamline maintenance, reduce downtime, and enable the creation of connected goods. The research is aimed at finding the digital representation of electro-mechanical railway interlocking elements are simulated and its functionality is verified accordingly. With respect to carry out this research, the Indian Railways Mangalagiri Railway Station blueprints were collected to represent the physical Model which consists of signals, points, a crank handle, an LC gate, and many more elements. In this paper, three interlocking elements such as points, crank handle, and LC gate, are analysed, modelled, and simulated by using Model Sim ME of Libero SoC software. Representation of these digital elements were developed using a smart design canvas. Therefore, the proposed methodology will enhance the safety measures and security levels of the train movements and reduces the time required to replace the faulty relay. Similarly, in future, the proposed model could be dumped into the FPGA kit to verify the real-time data for finding the faulty relay in the relay-based interlocking system.

1. Introduction

In the demanding era of industry 4.0, digital twins (DT) are used as a tool to connect physical and digital worlds. A virtual representation of a physical system, process, or product can be used for diagnostics and design adjustments is called a Digital Twin. Manufacturing companies are using the leverage of DT technology to reduce development costs of the next generation of machines by 50 %. Digital twins are being used by businesses to optimise their products in ways that were previously either impossible or unrealistic [1–3]. It serves as a link between the physical and digital worlds by using sensors to gather information in real-time situation about a physical object which can then be utilised to construct a digital replica of the object. It can receive inputs from sensors gathering data from a real-world counterpart, simulate the physical object in real-time, and provide insights in the form of performance and potential problems [4]. A digital twin can help to troubleshoot the potential faults which ultimately improves customer satisfaction. It also helps with product differentiation, product quality, and add-on services. Primarily, the digital twin is composed of physical entities in the real world, their virtual models and the connected data that tie the two worlds [5].

India is considered as one of the highest populated countries with dense population in the world. The major mode of public transportation in India is through the Indian Railway. Due communication system failures frequently accidents occur, may be due to

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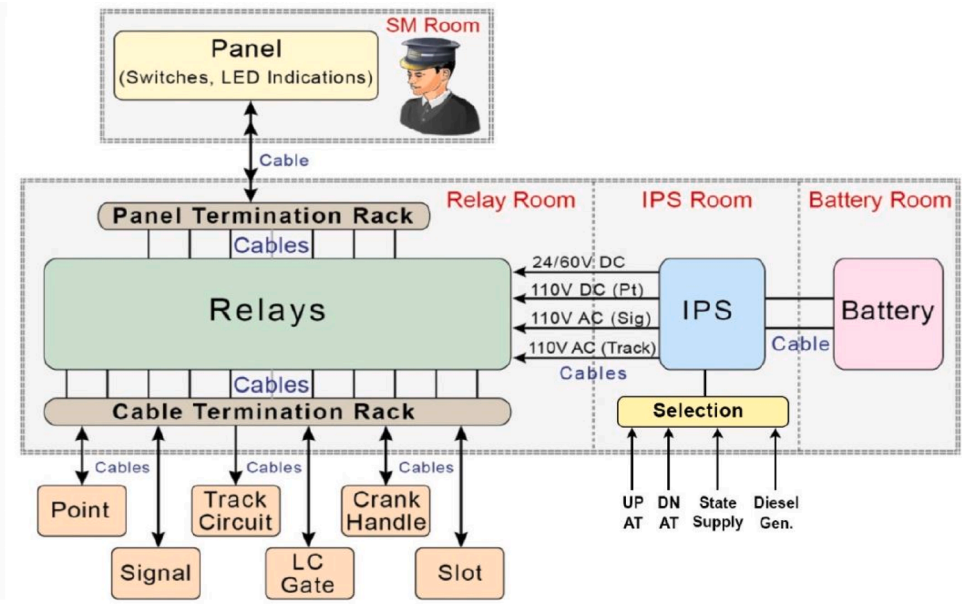


Fig. 1. A pictorial view of the Electro-Mechanical railway interlocking system plant that consists of different interlocking elements.

Table 1
Different stages of interlocking.

S. No	Name of the stage	Description
1	Route setting & Signal clearance	Non-route setting type panels - Here the route is set by an individual operation of points followed by signal operation. Route setting type panels - Route setting and signal clearance is done by a single operation by a selection of SIGNAL & ROUTE
2	Route checking	Points in the route are correctly set - conflicting routes are not set
3	Route locking	Points and LC gates are locked
4	Signal control	If the track is clear of other trains - The relay that controls the signal gets up
5	Signal lighting	The signal control relay cuts off the supply to red lamp & lights yellow/ green lamp
6	Route release by train movement	Route release takes place after checking sequential dropping & picking up of track circuits and changing the signal switch to normal
7	Route cancellation before train movement	If dead approach locking is provided - the first signal switch is normalized - then route cancellation is applied - after 2 min the route is released. If approach locking is provided and the train has not yet occupied the approach track - after normalization of a signal switch, the route also is released without time delay. If approach locking is provided with the information that and the train has occupied the approach track - after normalization of a signal switch, route cancellation is to be applied - after 2 min delay route is released
8	Route cancellation after train movement	After complete passage of the train - if all track circuits are clear - the signal switch is normalized and then cancellation is applied - after two minutes the route is released. After complete passage of the train - if the route is not released & a track circuit is down due to failure - the route is released by applying super emergency cancellation - some railway stations/crossings do not have this facility - the maintainer releases the route.

human and mechanical errors. Therefore, we need to innovate and evolve with smart devices and systems everywhere to automate the entire railway signalling system to handle greater amount railway passengers. Currently, the interlocking is used in the railway signalling system. It is defined as an arrangement of signals, points, and other appliances that are operated from a panel or a lever frame to avoid the collision and derailment of trains [6]. Interlocking can be done either mechanically, electrically, or both and the operations on signals and points must be carried out in the correct order to ensure safety [7–9]. The interlocking is designed in such a way that it is unable to display a signal unless the route is proven to be safe [10]. On the other hand, if a fault had occurred in the interlocking system, it must be capable to evolve itself to a state that provides safety to people. For example, due to a fault, the order is received that a traffic light must either be green or red at the same time, and then the interlocking system decides on the red aspect only [11,12]. An interlocking plant is a term that is to refer to both the signalling devices and the rails, as shown in Fig.1.

Initially, the commands for operation is given through switches/buttons on the panel boards located in the Station Master's (SM) room. Relay system is operated through the relay room process using commands, and if it is safe – the command is delivered to the field equipment like signal, point, level crossing (LC) gate, crank handle etc. The signalling system is powered by an integrated power supply (IPS) unit continuously. An interlocking system mainly consists of the following elements such as:

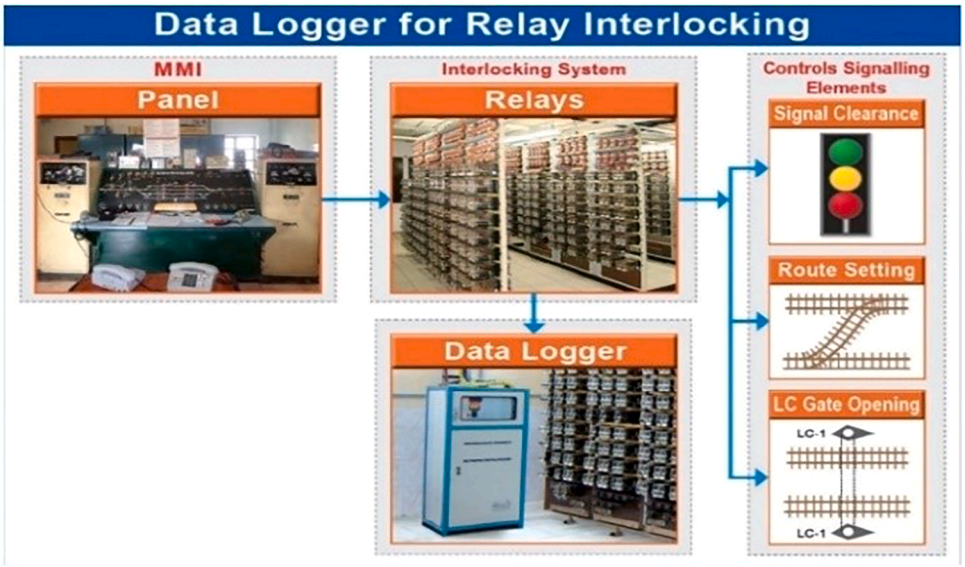
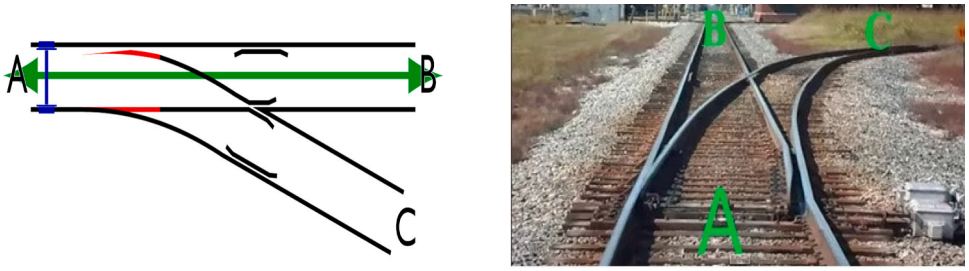


Fig. 2. Function description of Electro-Mechanical (Relay) Railway Interlocking System.



(a) - General representation of Point *(b) - Practical representation of the Point.*

Fig. 3. Point.

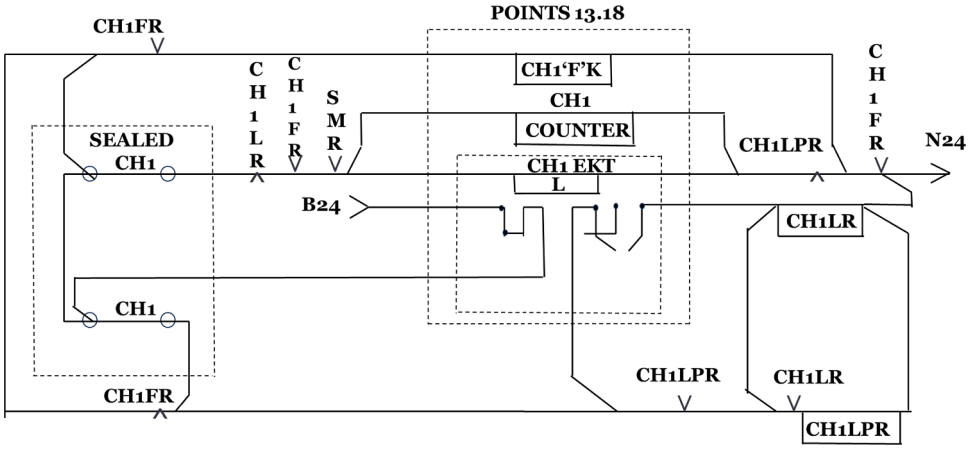


Fig. 4. Physical representation of the Crank Handle Locked Relay.

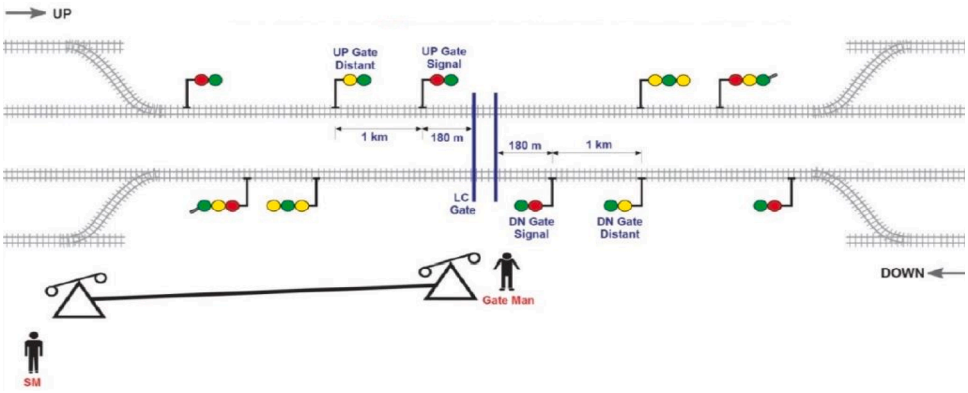


Fig. 5. Interlocking of LC Gate in Block Section.

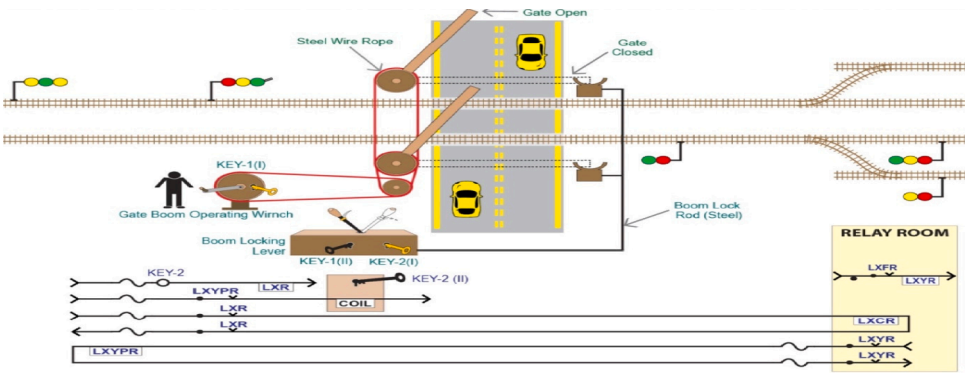


Fig. 6. Interlocking of LC Gate in Station Section that shows the Lifting Barrier Gate closed Status Transfer to SM-Gate and open permission transfer by SM to Gateman.

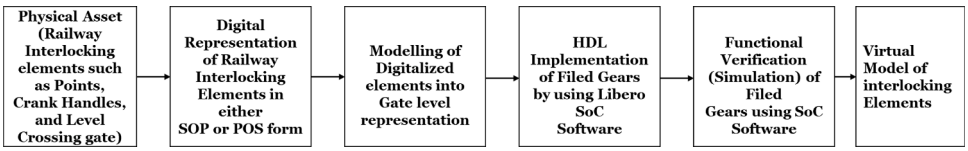
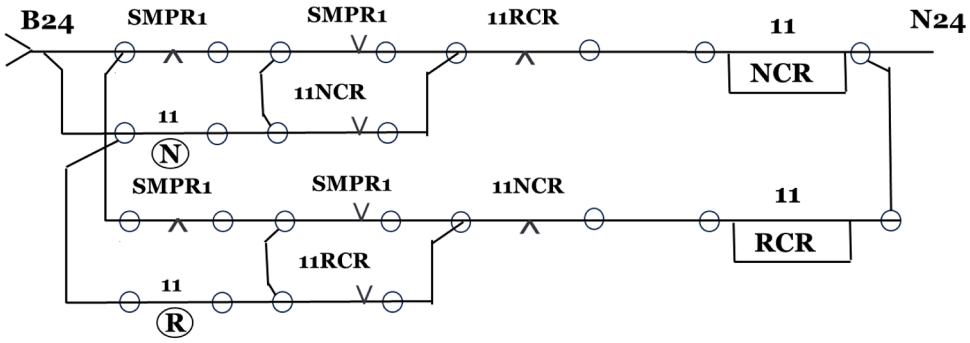
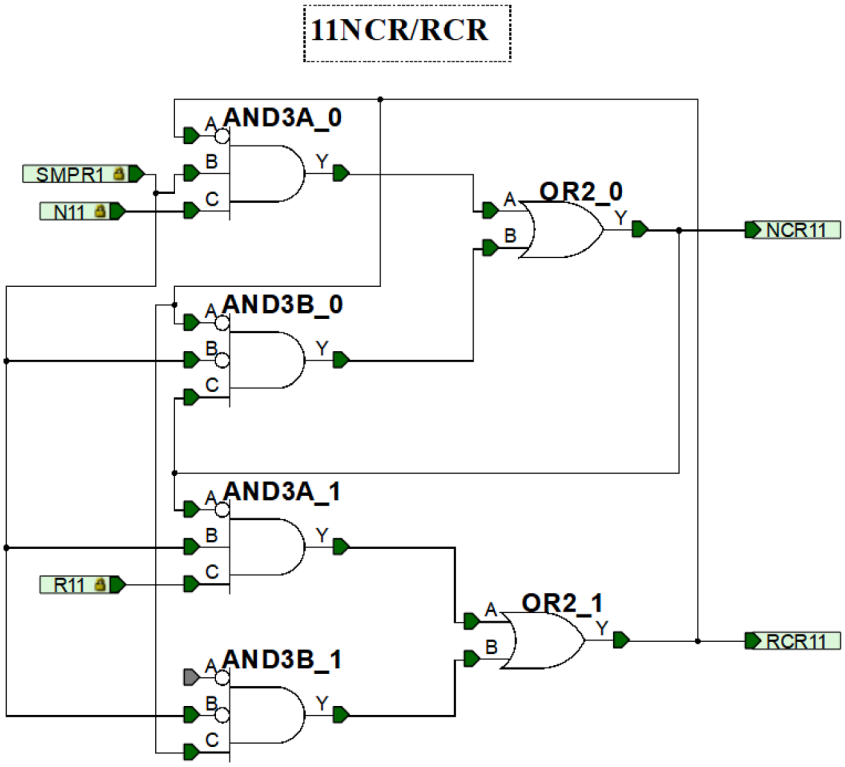


Fig. 7. Proposed Model to create the digital model representation of railway interlocking elements.

- Point: It is a track structure used to divert trains from one line to another line. It is operated by SM from a panel to set the route for a train [13].
- Signal: These are used as an indication to the train driver to enter the next section of the track while maintaining a specific speed. The signal element has different colours called aspects such as red to stop driving, yellow to drive with caution and reduced speed, and green to indicate normal speed [14].
- Track Circuit: These are the different sections in which the track is separated. Each of these sections is provided with a detector, called a track circuit which allows for detecting the presence of a train in the section [15,16].
- Level Crossing (LC) Gate: Each pedestrian or vehicular crossing on the railway line must be protected by a labyrinth with a bell, buzzer and a barrier known as the LC gate [17].
- Route: The term route is defined as the path that the train can take between two traffic lights. It is made up of one or more-track sections. The elements that intervene in each route and their status to be enabled are not described in the so-called "interlocking table".
- Crank handle (CH): When the point fails, the movement of rails can be performed manually through the crank handle.



(a) - The physical representation of the Point circuit diagram



(b) – Digital Modelling of Point element

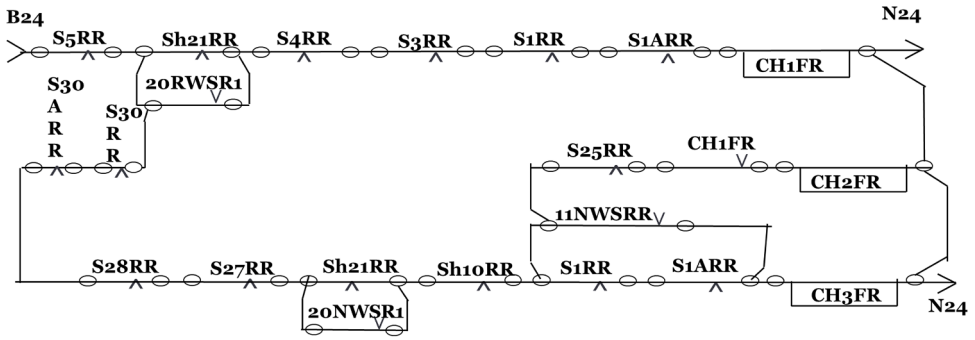
Fig. 8. Point in the interlocking system.

1.1. Stages of interlocking

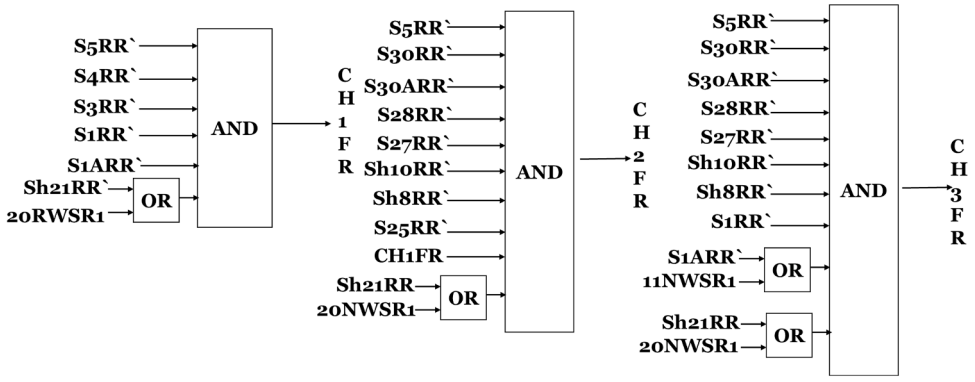
There are different stages of interlocking with variety of elements which are present in the railway signalling system as shown in Table 1.

In this research, the virtual modelling of interlocking, a component of the railway signalling system ensures the safety of Indian rail movements is considered [18–20]. The proposed system ensures high reliability as it is not labour-intensive and as such does not prone to any error.

Therefore, the main contributions of this research are as follows: (1) Identified the different railway interlocking elements along with their working principles. (2) Collected the physical circuit diagram of the Mangalagiri Railway Station for evaluation of the electro-mechanical interlocking system. (3) Digital representation of the interlocking elements using Boolean expressions in the form of SOP/POS by analysing the internal mechanism. (4) HDL implementation and functional verification of the Boolean Expressions using Libero SoC software. The different sections of the research paper are structured as follows - Section II reviews the basic working of different interlocking elements such as points, crank hand, and LC gate. And, Section III describes the digital representation of



(a) - The physical representation of CH free relay (CHFR),



(b) – Digital representation of CHFR

Fig. 9. Crank Handle.

interlocking elements along with their implementation details. The Section IV, elucidates the functional verification and simulation of interlocking elements with a detailed explanation by showing the results. Finally, Section V, concludes the research paper by explaining the future scope.

2. Railway interlocking system

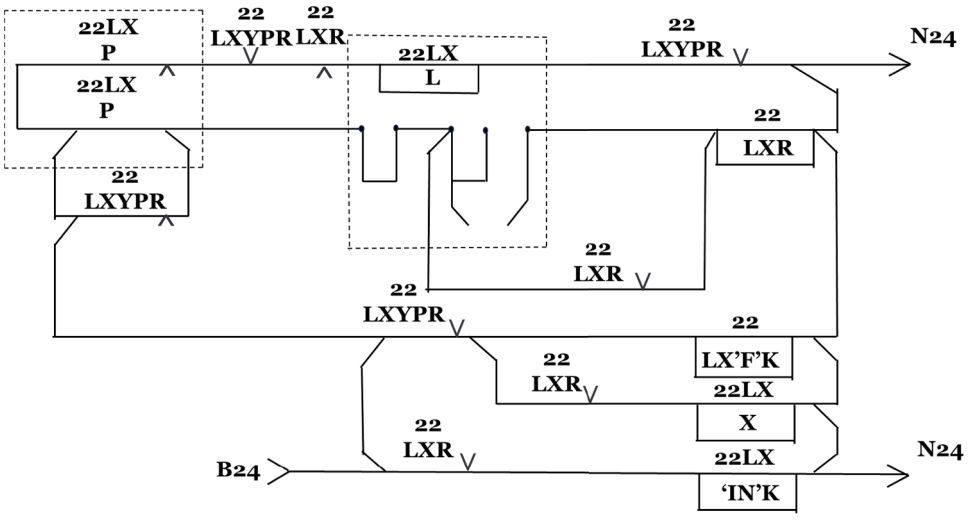
There are three kinds of railway interlocking systems present such as mechanical, electro-mechanical (relay) and computer-based systems [21]. In this research, the virtual modelling of a relay-based interlocking system was considered as a part of the implementation of the digital twin [21]. The Fig.2. represents the relay-based, metal-to-carbon type relay interlocking system including a VDU panel, relays, data logger and field gears. Whenever the SM wants to allow/dispatch a train into a station, he will operate the signal from the panel to meet the required operation based on the status of the relays and conditions of interlocking [22,23]. The status of relays stores in the data logger from time to time [24,25,14]. In the process of this research, digital modelling was carried out on route setting devices such as points, crank handle (CH) and level crossing (LC) gates. So, an elaborated working principle of these three elements were discussed in the following section.

2.1. Point

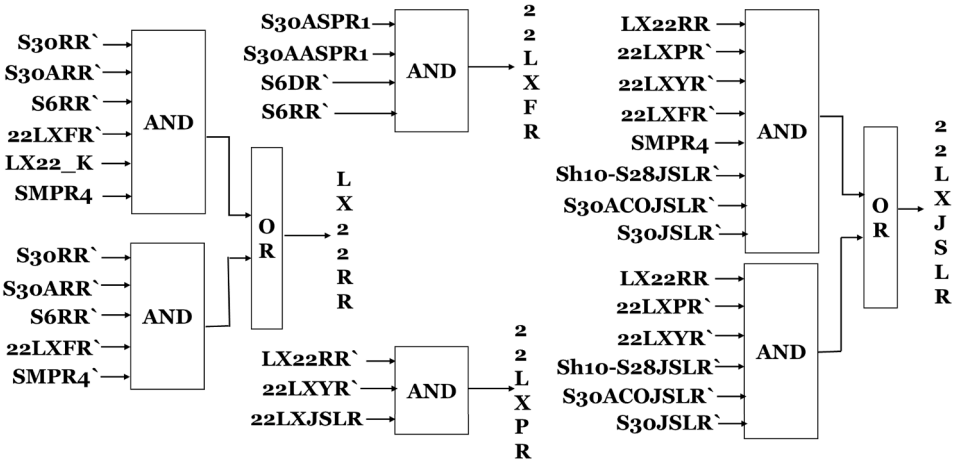
A point is a motor that allows the movable rails to either be normal or reverse that allows train to be either on the main track or crossing respectively [26]. Let's say a point is set to normal state, in which case the train goes from A to B; if the point is set to reverse, the train goes from A to C as shown in Fig 3. The general and practical representation of the element "point" is shown in Figures 3(a) and 3(b). Points can be operated in two ways such as rod operated, and motor-operated. In this research, motor-operated points are considered for implementation of the virtual model.

2.2. Crank handle

If the operation of a point fails due to any error, then the point can be operated via CH. After a signal has been set, manual operation of the point could affect the train's ability to run, therefore CH can be appropriately linked with signals. As there are several points in a station, giving CH interlocking for each point separately is not feasible [27]. Additionally, it is inappropriate to use a single crank



(a) - The physical representation of energized LXR relay



(b) – Digital representation of LC gate.

Fig. 10. LC Gate.

handle for all the points. So, to obtain maximum flexibility, points are grouped, and the concept of CH is included in interlocking elements as follows:

- The relevant CH, which is stored inside an electrical key transmitter (EKT), cannot be unlocked once a signal has been set across the points.
- When CH is removed, it is not permitted to operate the panel points along with its related signals.
- No other group point machine may accept the crank handle that was removed from another group.

To achieve the aforementioned interlocking, the crank handle must be linked and welded to the EKT key. The insertion of a key into the EKT and its subsequent clockwise twist causes the CH in locked relay (CHLR) to accelerate and sustain front contact, as seen in Fig. 4. The crank handle EKT is kept in the provided glass-fronted box secured with a padlock [28]. Each time the crank handle is released to manually operate the point, SM must enter data in the CH register.

2.3. Level crossing gate

The purpose of the LC gate is to ensure traffic movement across them in a controlled way and in accordance with the established guidelines [29]. The term level crossing is used as the level of incoming/ougoing road traffic and railroad tracks are crossing each

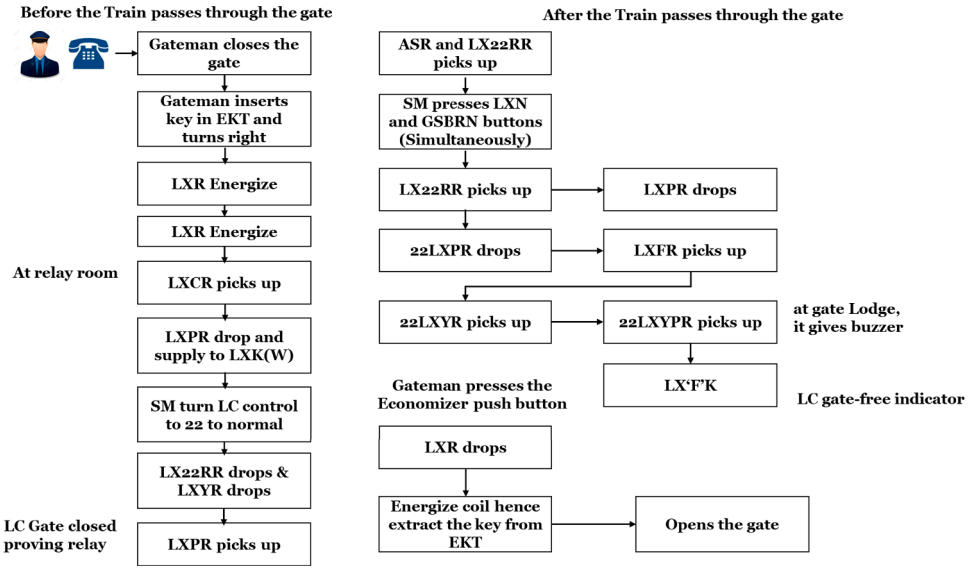


Fig. 11. The sequence of relays operation for opening and closing of an LC gate.

other. There are two different types of LC gate elements present as below:

LC Gate in block section – It is provided with a gate signal at 180 m from the gate. It displays green when the gate is closed and red when the gate is open. Behind the gate signal, gate distance is provided. It displays yellow when the gate is open and green when it is closed [30]. After closing the gate, the gateman operates the gate signal as shown in Fig. 5. [Without closing the gate, it is not possible to operate the gate signal- after operating the gate signal also it will not be possible to open the gate.]

LC Gate in station section – It is interlocked with a distant, home, starter or advanced starter depending upon its position. After closing the gate, the gateman electrically transmits gate control to SM [31]. Gate closed status is used in the interlocking by a signal called LC gate closed relay (LXCR). Unless the station master gives back the control electrically, the gateman cannot open the gate [32].

As shown in Fig. 6. Initially, the SM phones the gateman to request that the gate be closed. Next, the gatekeeper shuts the gate, locks the boom, takes out the key, places it inside EKT, and rotates it in a clockwise direction to transfer control. As a result, the LXR relay sticks through and energizes itself [33]. The gateman can therefore leave the key in the EKT once LXR has been picked up. Without SM's assistance, Gateman has no control over the LC gate's opening.

3. Implementation details

The main aim of this research is to evaluate the performance of digital representation of interlocking elements.

3.1. Proposed methodology workflow

To evaluate the proposed model, Mangalagiri Railway Station interlocking circuit diagrams were collected. Initially, the working principle of these circuits was analyzed by a thorough study of interlocking elements. Based on the operation, the elements were realized into digital form from electro-mechanical. Now, these digital elements are modelled using logical gates and implemented in a software called Libero SoC. This allows the use of multiple design entry techniques, such as schematic-based design and Hardware Description Language (HDL) entry utilizing Verilog or VHDL. This gives designers the freedom to select the approach that best fits their needs and workflow. Its workflow and interface is set up to make the design process easier for both inexperienced and seasoned FPGA designers. Libero SoC typically comes with collaborative design features that let numerous engineers collaborate on different parts of a project at the same time. To streamline and organize the design process, it might also offer project management features. The physical equivalent digital element function is verified to obtain the digital twin interlocking elements. The total methodology of the proposed model is shown in Fig. 7.

3.2. Modelling of interlocking elements

In this paper, the HDL implementation of the points, crank handle, and LC gate is designed using Libero SoC software in the canvas.

3.2.1. Point modelling

Figs. 8(a) and 8 (b) show the physical and DT representation of a point element of a single-point machine. In this 11 is a point, and it contains two connections i.e., Normal and Reverse.

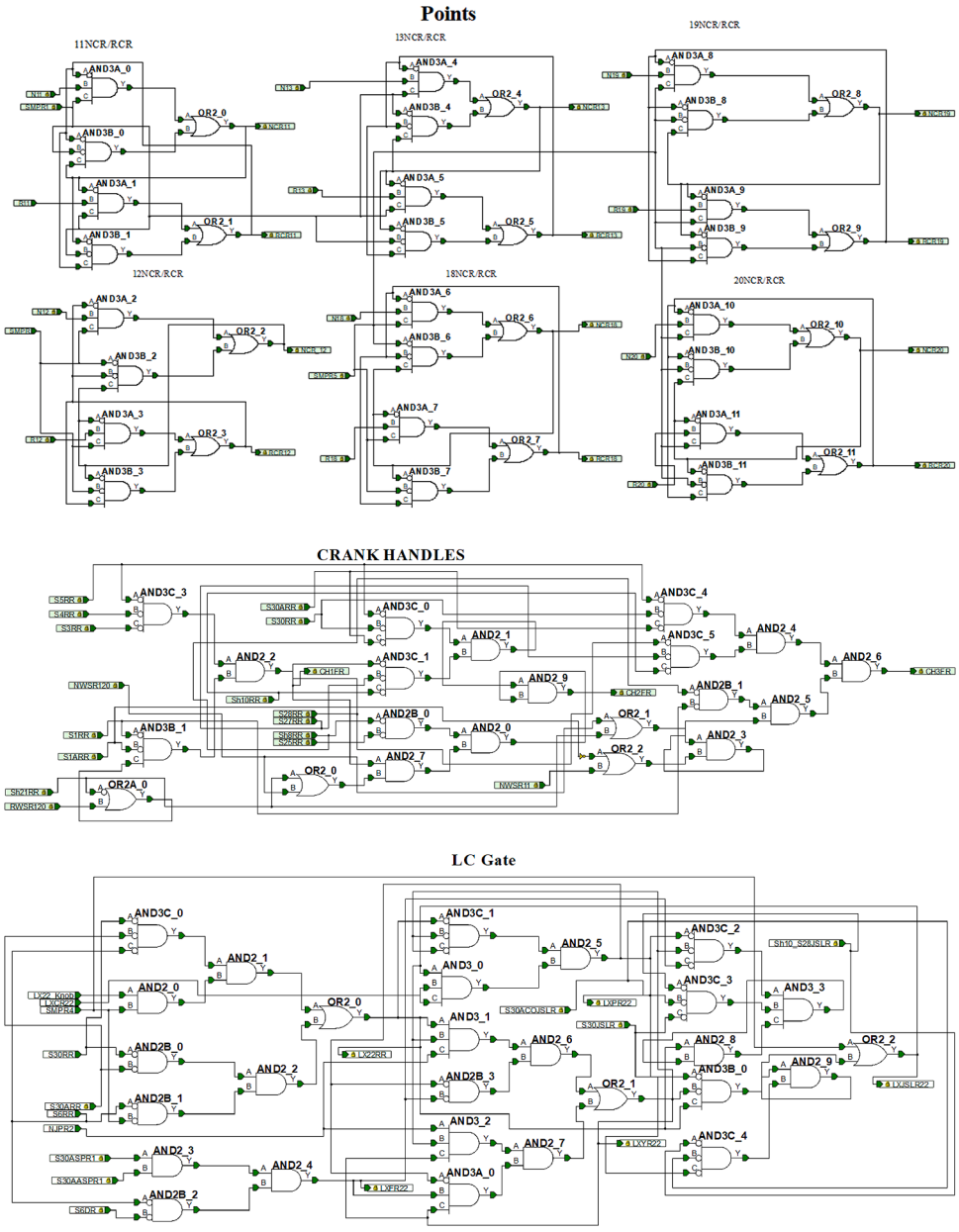


Fig. 12. Digital Model of Interlocking System with Points, CHLR and LC gate in a single unit.

If the point is connected to normal, then normal contact relays (NCR) will be picked up. If it is connected to reverse, then reverse contact relays (RCR) is picked up. Here, the logic for picking up of NCR is that RCR should drop its position and the SMPR should be present. Similarly, for picking up RCR, the NCR should be in a drop position otherwise it will not be picked up. This entire operation is performed in the presence of SM only which is proven by the SMPR relay. On the other hand, if the SM left the room, the total operation will be locked at the previous state only. This entire working of the point element is represented in the digital form. The Mangalagiri Railway Station consists of six points that can work on similar logic, as an example, point 11 is considered as shown in Eqs. (1) and 2. The other points are also realized similarly.

$$11NCR=N11.SMPR1.!11RCR+!SMPR1.11NCR.!11RCR \quad (1)$$

$$11RCR=R11.SMPR1.!11NCR+!SMPR1.11RCR.!11NCR \quad (2)$$



Fig. 13. (a) - Point 11 Normal where SM key is present (b) - Point 11 Reverse where SM key is present (c) - Point 11 Reverse where SM key not present.

3.2.2. Crank handle modelling

When an electrical point fails, crank handles are available to manually operate motor-operated points. These are arranged in the CH zone according to working convenience. Signals are interlocked with crank handles. CH cannot be removed while signals are turned "off," and vice versa. Figs. 9(a) and 9 (b) show the physical and DT of a CH element. Mangalagiri Railway Station consists of six points which are grouped into three elements named CH1FR, CH2FR, and CH3FR respectively.

CH1FR will be dropped when a signal is on. At that point, CH1LR does not drop if SM tries to withdraw the crank handle by pushing the economizer push button since the feed extended to CH1LR through CH1FR back contact, as seen in Fig. 4. The CH key cannot be withdrawn as a result, and the EKT lock coil isn't energised. The signals that are sent off continue to exist in the same state at the same time. CH1FR stays energized if a signal is not shut off. CHLR decreases when SM hits the push button for the economizer. The key is removed and the lock coil is energized. As a result, the signals that are interlocked with it cannot be removed. The CHLR circuit is built in such a way that when the economizer push button hits to remove the crank handle, it cuts off power to CHLR and CHLR drops, ensuring that CHLR drops before the actual extraction of the key. Crank handle extraction is not possible till CHLR drops.

For implementing the crank handle, need to know about the relay called CHFR. If this relay is free, then the CH is used for setting the points. CH1FR will be in an energised state when all the relevant signal dials are normal, meaning that the RRs have been dropped. After that, the CH box's crank handle free indicator will be lit. As a result, it lets SM know that CH1 can be removed if necessary. The total working of the three CH elements is represented in the digital form as shown in Eqs. (3), 4, and 5

$$CH1FR=!\ S5RR.\ (!\ Sh21RR+20RWSR1).!\ S4RR.!\ S3RR.!\ S1RR.!\ S1ARR \tag{3}$$

$$CH2FR=!\ S5RR.!\ S30RR.!\ S30ARR.\ !S28RR.\ !S27RR(Sh21RR+20NWSR1)!Sh10RR.\ !Sh8RR.!S25RR.CH1FR \tag{4}$$

$$CH3FR=!S5RR.!S30RR.!S30ARR.!S28RR.!S27RR.(!Sh21RR+20NWSR1)!Sh10RR.!Sh8RR.!S1RR(!S1ARR+11NWSR) \tag{5}$$

Table 2
Tabular form Point Simulation.

Point	N18	R18	SMPR1
18NCR	↑	↓	↑
	↑	↓	↓
18RCR	↓	↑	↑
	↓	↑	↓
19NCR	N19	R19	SMPR5
	↑	↓	↑
	↑	↓	↓
19RCR	↓	↑	↑
	↓	↑	↓
20NCR	N20	R20	SMPR5
	↑	↓	↑
	↑	↓	↓
20RCR	↓	↑	↑
	↓	↑	↓
Point	N11	R11	SMPR1
11NCR	↑	↓	↑
	↑	↓	↓
11RCR	↓	↑	↑
	↓	↑	↓
12NCR	N12	R12	SMPR
	↑	↓	↑
	↑	↓	↓
12RCR	↓	↑	↑
	↓	↑	↓
13NCR	N13	R13	SMPR1
	↑	↓	↑
	↑	↓	↓
13RCR	↓	↑	↑
	↓	↑	↓

3.2.3. Level crossing (LC) gate modelling

Whenever a train passes through a gate, the SM communicates the gateman to close the gate over phone. Then the gateman closes the gate by inserting a key in EKT and turning it in a clockwise direction which leads to the energization of the LXR relay as shown in Fig. 10(a). If the LXR relay picks up its corresponding LXCR relay will be energized in the relay room, by giving supply of LXK(W) to flash continuously and drops the LXPR relay. Then, SM turns the LX22RR to a normal position which leads drop of two relays LX22RR, LXYR, and pick up of LXPR simultaneously to indicate the LC gate is closed by steadying the LXK(W) flash indication. After the train passes through the gate, the approach sticks relay (ASR) picks up automatically, SM presses the LXN and GSBRN button simultaneously which gives a buzzer at the gate. Also, the LC gate-free indicator (LX'F'K) is picked up, thereby the gateman opens the gate. The total working of the LC gate elements are represented in the digital form as shown in Equations from 6 to 10. From these Boolean equations, the digital twin of an LC Gate is modelled as shown in Figure 10(b). The entire operation is shown in Fig. 11 through a flow chart.

$$LX22RR=!S30RR.!S30ARR.!S6RR.LX22R_b.SMPR4+!S30RR.!S30ARR.!S6RR.!SMPR4.LX22RR \quad (6)$$

$$22LXYR=LX22RR.!22LXPR.!22LXFR.22LXJSLR.NJPR2+LX22RR.!22LXPR.22LXFR.\ 22LXYR.22LXJSLR.NJPR2 \quad (7)$$

$$22LXFR=S30ASPR1.S30AASPR1.!S6DR.!S6RR \quad (8)$$

$$22LXPR=!LX22RR.!22LXYR.22LXJSLR.22LXCR.!22LXYR.22LXCR \quad (9)$$

$$22LXJSLR=LX22RR.!22LXPR.!22LXYR.!22LXFR.SMPR4.!NJPR2.CANCELBUTTON.LX22RR.!22LXPR.!Sh10S28JSLR.!S30AC0JSLR.!S30JSLR+LX22RR.!22LXPR.!22LXYR.22LXJSLR.LX22RR.!22LXPR.!Sh10- S28JSLR.S30AC0JSLR.S30JSLR \quad (10)$$

3.3. DT of railway interlocking elements

Fig. 12 shows the implementation of points, crank handle, and LC gate as per the logical circuit diagram. This is implemented in Libero SoC of smart design by taking up necessary components in smart canvas and simulated. Libero SoC Design Suite's broad features enable excellent productivity and provide access to a comprehensive set of hardware tools, including development kits, a broad IP library, and RTL entry through programming. All the individual elements are described and how they are implemented is explained in the above content. In these, all three interlocking elements are designed within a single smart design canvas as per the Boolean expression. Here, stimulus is applied in such a way that whenever SM presses the knob/button that signal gives a high value i.e., one and the necessary signals are also given a high value. The simulation results are shown in the next phase.

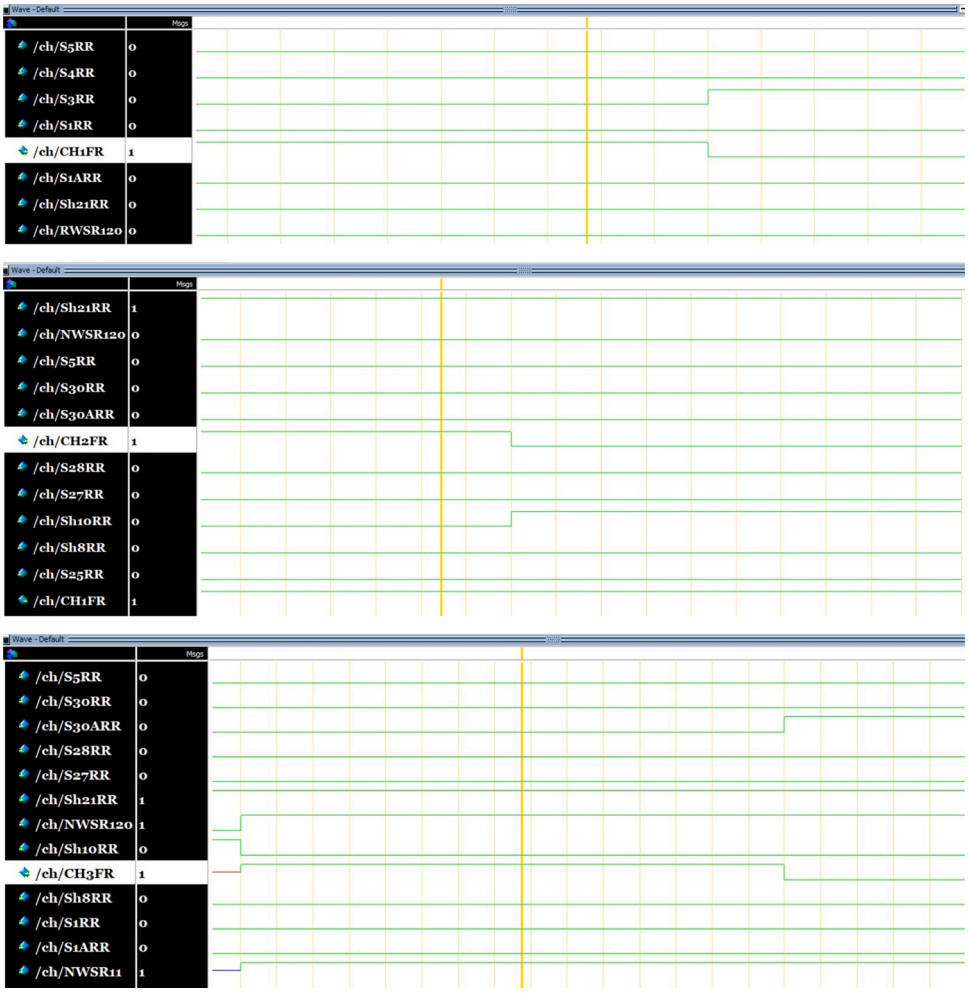


Fig. 14. Three different Crack Handle Simulations.

Table 3
Tabular form of Crank Handle Simulation.

CH1FR	S5RR ↓	S4RR ↓	S3RR ↓	S1RR ↓	S1ARR ↓	Sh21RR OR 20RWSR1 ↓				
CH2FR	S5RR ↓	S30RR ↓	S30ARR ↓	S28RR ↓	S27RR ↓	Sh10RR ↓	Sh8RR ↓	S25RR ↓	CH1FR ↑	Sh21RR OR 20NWSR1 ↑
CH3FR	S5RR ↓	S30RR ↓	S30ARR ↓	S28RR ↓	S27RR ↓	Sh10RR ↓	Sh8RR ↓	S1RR ↓	S1ARR OR 11NWSR ↑	Sh21RR OR 20NWSR1 ↑

4. Simulation results and discussions

In this part of the research, all three interlocking elements are simulated for functional verification using the Model Sim ME simulator in libero SoC. This simulator is very simple to use with the help of constantly updated panes, a graphical user interface enables you to quickly detect and debug problems. For instance, the source, signals, process, and variables windows are automatically updated when a design region is selected in the structure window. The debug environment is made simple to use by these cross-linked Model Sim windows. You don't need to exit the simulator to change, recompile, or re-simulate after an issue is identified. The simulated results are shown below. To ensure the working condition of a point, an example contact relay 11 is considered. To set the point relay in the normal position, the SM should press N11 knob which energizes NCR11 relay. To achieve this, the signal N11 value is given high as shown in Fig. 13(a) and its corresponding NCR11 value becomes high. On the other hand, to set the point relay in reverse position, the SM should press R11 knob which energizes the RCR11 relay. To achieve this, the signal R11 value is given high as shown in Fig. 13(b) and its corresponding RCR11 value becomes high. In both the cases, the value of SMPR should give logic high to represent the presence of SM. If the SM key is removed from the panel, the operation remains as it is i.e., the previous state only. To verify this

Table 4
To simulate the working of an LC gate, initially, the knowledge of relays that are involved in interlocking is to be known.

Indication and actions at Panel	Actions at LC Gate
	Gate man closes the Gate – takes out the gate key from the boom lock - inserts it EKT, LXR relay picks up and the key gets locked in the EKT
LXCR up in station relay room by LXR at LC	
LC closed indication flashes on the panel	
SM operates gate switch- LXRR DN	
Gate closed indication becomes steady - LXPR Up	
Signal is taken off – LXFR DN	
The train passed and the route released – LXFR Up	
Gate switch operated – LXRR Up + LXFR Up = LXYR Up Gate open indication flashes	LXYPR Up at LC gate
	Gate man removes the key from EKT — LXR drops
LXCR drops Gate open indication becomes steady	

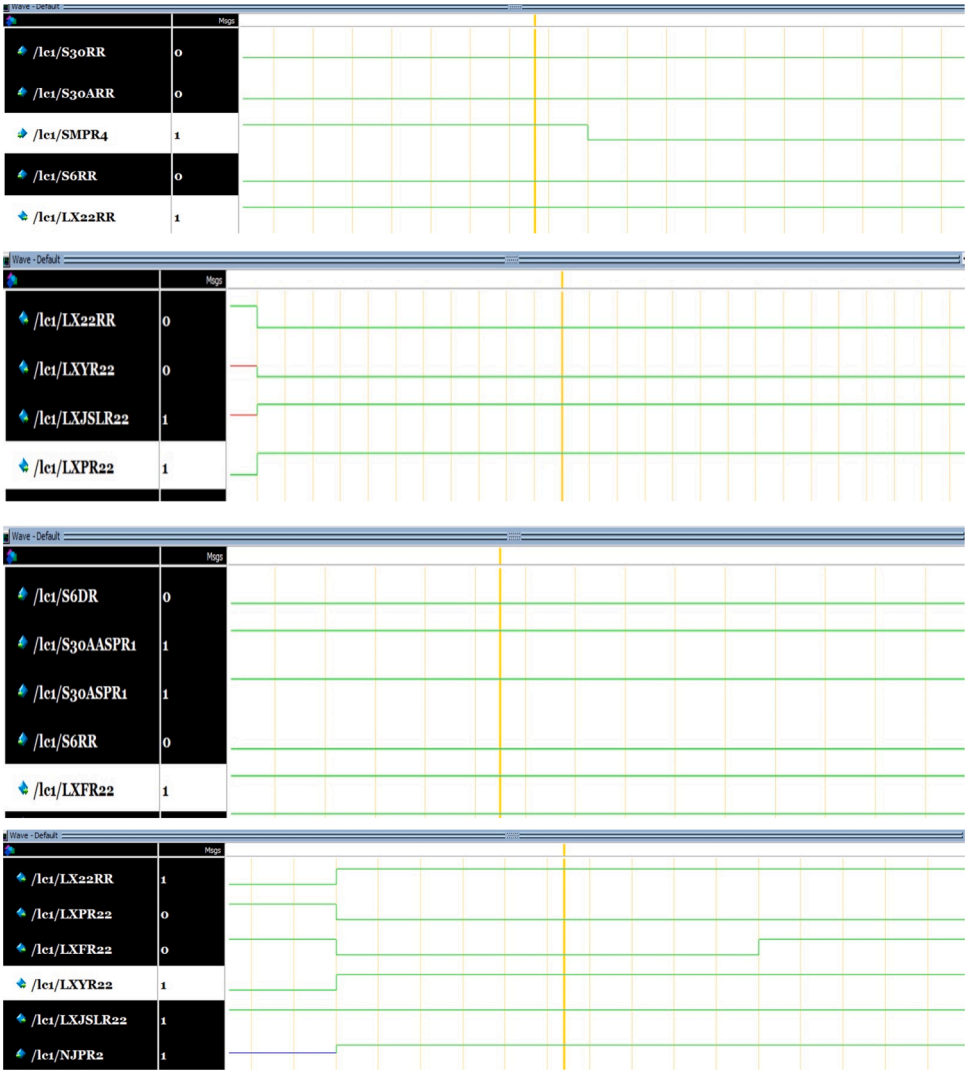


Fig. 15. Level Crossing Gate Simulation.

logic, SMPR is given to logic zero, the panel is in its last operation i.e., RCR11 is picked up as shown in Fig. 13(c). As already mentioned, Mangalagiri Railway Station contains a total 6 number of points, all those points are modelled, analyzed, implemented, and simulated.

Table 2. shows the tabular form of all point simulations for achieving a particular point, and table 2 shows the status of relays for achieving particular point. To understand the working of a crank handle, CH1FR will be discarded if any of the involved signals are

Table 5
Tabular form of Level Crossing Gate Simulation.

LX22RR	S30RR	S30ARR	S6RR	LX22_knob	SMPR4			
	↓	↓	↓	↑	↑			
	↓	↓	↓	↑	↓			
22LXFR	S30ASPR1	S0AASPR1	S6DR	S6RR				
	↑	↑	↓	↓				
22LXPR	LX22RR	22LXYR	22LXJSLR	22LXCR				
	↓	↓	↑	↑				
	LX22RR	22LXPR	22LXFR	22LXJSLR	NJPR2			
22LXYR	↑	↓	↓	↑	↑			
	↑	↓	↑	↑	↑			
	LX22RR	22LXPR	22LXYR	22LXFR	SMPR4	Sh10-S28JSLR	S30ACOJSLR	S30JSLR
22LXJSLR	↑	↓	↓	↓	↑	↓	↓	↓
	↑	↓	↓	↓	–	↓	↓	↓

taken off, i.e., RR picked up. When the crank handle free indicator goes out, SM realises that the signal needs to be normalised before CH can be extracted. So, all the signals should be taken off which is shown in Fig. 14. When SM pushes the economizer push button, CHLR will be set to zero. Then the key is withdrawn from it when the lock coil is powered up. As a result, the signals that are interlocked with it cannot be removed.

The design of the CHLR circuit ensures that CHLR lowers before the key is really extracted. When the economizer push button is pressed to remove the crank handle, CHLR loses power.Until CHLR drops, crank handle extraction is not possible. Handle simulation cracks CHFR will be discarded when a signal is removed. As the feed is extended to CHLR through the CHFR back contact, as seen in Fig. 4, CHLR does not drop if SM tries to extract the crank handle at that point of time by pressing the economizer push button. As a result, the crank handle key cannot be withdrawn and the EKT lock coil remains deactivated. In addition, the signals received, as already mentioned, Mangalagiri Railway Station contains a total of 3 CH, the simulation for three individual CH is shown in Fig. 14.

For operating any crank handle the associated relays should be in a particular position which is shown in Table 3. It also represents the simulation results of the crank handle. Whenever the LC gate is operated there will be a sequence of relays will be picked up and dropped parallelly as shown in Table 4. During the LC gate operation, the concept of interlocking is applied to specific signals which are enabled by the SM by pressing its corresponding buttons. For example, if SM wants to operate LX22R button to a particular position, there will be a set of conditions to be proved. Then only it is possible to operate the specific button. These set of conditions are interlocked in such a way that if any of the conditions are not satisfied then LX22R fails to operate. These specific conditions are already mentioned in Eqs. (6) to 10, and the simulation results are shown in Fig. 15.

For LC gate simulation, some relays should be in a turn off position and some relays in turn on which is shown in Table5. Fig. 15 shows the all-necessary conditions for operating the LC gate. Those conditions are listed in the Boolean Eqs. (6)-10. By observing the waveform, it is said that all associated signals should be turned off for operating LC gate. One more observation is that all the relays are associated with a numerical value 22 because these relays belong to the LC gate number 22, that's why all the knobs and relays start with the numerical value 22. For operating LX 22RR button by SM, the associated signals S30RR, S30ARR, and S6RR signal relays should be in the off position which is shown in Fig. 15 and the SMPR relay also in on position that is shown in Fig. 15. If the train passes through the LC gate, then the LC gate should be open for this purpose whenever the train crosses through LC gate the S30ASPR1, and S30AASPR1 relays will be picked up and it is shown in Fig. 15.

5. Conclusion & future scope

In this paper, the importance of Digital Twin concept in Industry 4.0 is discussed. For creating a digital twin, the physical asset should be modelled, simulated and it should be implemented. As a part of creating a digital twin of railway interlocking system, the interlocking elements such as points, crank handles, and LC gates are modelled and simulated. By the above evidence of waveforms, it can be said that the modelling and simulation were done. Later the modelling of signals and corresponding elements has to be modelled and it should be simulated and implemented on an FPGA board which shows the virtual asset of a physical interlocking system. It should connect the physical system and virtual system by using interface protocol UART, which helps creating a digital twin for the railway's entire interlocking system. Particularly in India, these initiatives are still in the exploratory stage and have not yet been carried out realistically. As a result, we took time to conduct research it thoroughly and develop a virtual model. According to data available on the previous train accidents, our current system needs to be automated in a way that is more effective, accurate, and fail-safe.

We conclude that a Digital Twin-based system would prove to be more efficient in the current relay-based technology is used in India because the entire regulatory action would become automated. This would lower the time required to substitute the faulty relay efficiently and affordably.

In future work, the implementation of circuit diagrams could be dumped into the FPGA board. Which replicates the virtual asset of a physical system, whatever input data is given to the physical system, that data is stored in the data logger. So, the same data is given as an input to FPGA board, if any mismatches occur between the FPGA and physical asset, it displays the relay name. Suppose, when SM gives a red signal but the red light is not lit due to some faulty relay that should be on, but it is in off, the same data will be given to the FPGA board, there it occurs a mismatch of the relay, we will figure out that relay has to be replaced with a new relay. In this way, it

will be useful to improve operational efficacy, safety, and maintenance to verify the real-time data to find the faulty relay in the relay-based interlocking system. The expected outcome is to find the faulty relay among lakhs of relays, here the main challenges that might happen due to interface physical and virtual asset, for achieving this one, we have to develop a UART protocol for interfacing FPGA itself, the modelling part will be PL section and UART should be PS for transferring the real-time data to virtual asset.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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